

# DONI VISHWA VARDHAN

AI/ML Engineer

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## SUMMARY

AI/ML Engineer who's built and shipped RAG agents, MCP tool servers, and fine-tuned LLMs (LangChain/LangGraph, Qdrant, Ollama, QLoRA). IIT Delhi grad blending hands-on GenAI engineering with product ownership — Figma design, analytics, and end-to-end feature delivery.

## ACADEMIC DETAILS

### Indian Institute of Technology Delhi

2026

B.Tech in Biochemical Engineering & Biotechnology — CGPA: 7

## EXPERIENCE

### Management Trainee – Program Manager | Basic Enterprises

June 2026 – Present

*Products: CleverTap • Agent Flash • Invoice Rejection Tracker • Agent Retention Strategies*

- Designed, built, and deployed a Retrieval-Augmented Generation (RAG) AI agent for Agent Support and Call Center operations, taking it from prototype to internal production use.
- Led end-to-end product development, from Figma wireframes and requirement scoping to feature delivery, using data-driven insights to solve user problems.
- Analyzed CleverTap product analytics to identify user drop-offs and pain points, prioritized the roadmap, and collaborated with engineering and operations teams to define success metrics and deliver data-informed solutions.

## PROJECTS

### Support Triage Agent with MCP | [GitHub](#)

*Model Context Protocol (MCP), Ollama (local open-source LLM), scikit-learn (TF-IDF), Python, FastAPI*

- Built an MCP server exposing support-ticket triage as tools (list, retrieve, update, classify), plus a FastAPI backend and browser dashboard on the same tools — giving agents a working product UI, not just a CLI.
- Ran ticket classification fully locally via Ollama (Llama 3.2), with TF-IDF-based retrieval of similar past tickets grounding each classification (RAG-style) at zero external API cost.
- Built a full pytest suite (23 tests) with GitHub Actions CI, and evaluated the system with a custom harness (routing accuracy + LLM as-judge), achieving 100% routing accuracy and a 4.4/5 response-quality score

### Financial Intelligence Agent | [GitHub](#)

*Python, LangGraph, Qdrant, PostgreSQL, FastAPI, Docker, Ollama*

- Built a RAG-based financial research agent to analyze annual reports and financial documents using open-source LLMs (via Ollama), LangGraph, and Qdrant vector store with hybrid dense+BM25 retrieval.
- Developed tool-using agent workflows (PostgreSQL text-to-SQL, financial calculations, document retrieval) exposed as APIs via FastAPI, with plan-then-execute orchestration and sqlglot-validated SQL execution.
- Implemented an LLM-as-judge evaluation harness scoring answer correctness, groundedness, and routing accuracy, with a defense-in-depth SQL safety layer (validated + read-only) and Dockerized deployment.

### HR Policy Q&A — QLoRA Fine-Tuning | [GitHub](#)

*Qwen2.5-1.5B, PEFT (LoRA), bitsandbytes (QLoRA), TRL, Hugging Face Transformers*

- Fine-tuned Qwen2.5-1.5B-Instruct with QLoRA (4-bit NF4 quantization + LoRA adapters) on a company HR policy dataset, reducing trainable parameters to a small fraction of the full model.
- Achieved a BERTScore of 0.88 on a held-out evaluation set.

### Credit Card Fraud Detection

*Python, Scikit-learn, Imbalanced-learn*

- Ran EDA and applied SMOTE oversampling to correct severe class imbalance in transaction data.
- Achieved 0.93 ROC-AUC combining logistic regression, KNN, and decision trees; cross-validated with isolation forest and local outlier factor anomaly detectors.

## TECHNICAL SKILLS

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**Languages:** Python, SQL, Java, JavaScript, Dart

**ML / Deep Learning:** Scikit-learn, PyTorch, TensorFlow, Keras, XGBoost, Hugging Face Transformers

**GenAI & LLM Engineering:** LangChain, LangGraph, RAG pipelines, Prompt Engineering, Vector DBs (Chroma, FAISS, Qdrant), Ollama, Fine-tuning (LoRA/QLoRA, PEFT, trl, bitsandbytes), Guardrails (Llama Guard), Gateways (LiteLLM), Model Context Protocol (MCP)

**MLOps & Deployment:** FastAPI, Docker, Kubernetes, MLflow, Azure ML, AWS SageMaker, Vertex AI, CI/CD (GitHub Actions), Data Bricks, React, Vercel, Next JS

**Data Engineering:** Pandas, NumPy, SQL

**Cloud:** Microsoft Azure, AWS, GCP

**Product & Analytics:** Figma, Product Analytics (CleverTap, Mixpanel)

**Tools:** Git, VS Code, MS Excel, MS Office